

Site Address:
Stanwell Power Station – Switchyard Rd, Stanwell
NCC 2022 Part J6-J9 Deems to Satisfy Report

Residential
Assessment

Commercial
Assessment

Sustainable
Consulting

Performance
Solutions

Daylight
Analysis

Moisture
Analysis

NCC 2022 Part J6-J9 Deems to Satisfy Report
Project Description: Commercial Refurbishments
Including Registry, Workshop & the Main Warehouse
Client: Deign Elements
Date: 25/03/2026

This report was completed by:

Matthew Morelli
NRG efficient homes
ESD Consultant
DMN Accredited Thermal Performance Assessor
BESS trained professional
Dip. Building Design (Architectural)



DISCLAIMER Although great care has been taken to prepare this report, NRGEH does not make any representations or give any warranties or assurances as to the accuracy of the information contained in the Report or whether the report is free from errors or omissions
NRGEH, its employees and agents shall not be liable for any loss, damage, cost or expense whether direct, indirect or consequential, incurred by, or arising by reason of, any person using or relying on the report and whether caused by reason or error, negligent act, omission or mis representation in the Report or otherwise. The Report has been prepared by NRGEH based on the information supplied.

© Copyright This plan shall remain the sole property of NRGEH and must not be given, lent, hired out or otherwise disposed of or copied without the written permission of NRGEH.

Table of Contents:

INTRODUCTION.....	4
PROJECT DETAILS:	4
SUMMARY OF REQUIREMENTS:.....	4
PART J6 - AIR-CONDITIONING AND VENTILATION SYSTEMS.....	5
PART J7 - ARTIFICIAL LIGHTING AND POWER	7
PART J9 - ENERGY MONITORING AND ON-SITE DISTRIBUTED ENERGY RESOURCES	9
CONCLUSION.....	10
APPENDIX.....	11

Introduction

The purpose of this report is to assess the proposed developments against Part J7 of the 2022 NCC Volume 1 and show compliance.

It has been deemed unnecessary to assess the building thermal envelope as the proposed works include either,

- the internal fit out of an existing building,
- a storage area,
- an area designed for manufacturing purpose and/or
- is a non-conditioned space.

Project Details:

Climate Zone: 2

Building Class: Class 5

This report is based on the following architectural plans provided by the client,

Project No: 25_003 Revision: P2 Date: Jan 2025

Summary of requirements:

Air-conditioning	No design specified at the time of assessment.
Artificial lighting	The aggregate design illumination power loads must not exceed Registry - Total 1594 watts Workshop - Total 2813 watts Main Warehouse - Total 1273 watts
Heated water supply	A heated water supply system for food preparation and sanitary purposes must be designed and installed in accordance with NCC Volume Three — Plumbing Code of Australia.
Facilities for energy monitoring	Not applicable

Part J6 - Air-conditioning and ventilation systems

NOTE: Air-conditioning and ventilation systems and components will be designed in accordance with the DTS requirements of Part J6 and a separate report will be submitted by the mechanical services engineer/designer to verify compliance where required.

	Required
J6D3 - Air-conditioning system control	An air-conditioning unit or system must: - must be capable of being deactivated when the building or part of a building served by that system is not occupied; and - when serving more than one <i>air-conditioning</i> zone or area with different heating or cooling needs, must—thermostatically control the temperature of each zone or area; and - not control the temperature by mixing actively heated air and actively cooled air; and - limit reheating to not more than— - for a fixed supply air rate, a 7.5 K rise in temperature; and - for a variable supply air rate, a 7.5 K rise in temperature at the nominal supply air rate but increased or decreased at the same rate that the supply air rate is respectively decreased or increased; and - which provides the <i>required</i> mechanical ventilation, other than in <i>climate zone</i> 1 or where dehumidification control is needed, must have an <i>outdoor air economy cycle</i> if the total air flow rate of any airside component of an <i>air-conditioning</i> system is greater than or equal to the figures in Table J6D3; and - which contains more than one water heater, chiller or coil, must be capable of stopping the flow of water to those not operating; and - with an airflow of more than 1000 L/s, must have a variable speed fan when its supply air quantity is capable of being varied; and - when serving a <i>sole-occupancy unit</i> in a Class 3 building, must not operate when any external door of the <i>sole-occupancy unit</i> that opens to a balcony or the like, is open for more than one minute; and - must have the ability to use direct signals from the control components responsible for the delivery of comfort conditions in the building to regulate the operation of central plant; and - must have a control dead band of not less than 2°C, except where a smaller range is <i>required</i> for specialized applications; and - must be provided with balancing dampers and balancing valves that ensure the maximum design air or fluid flow is achieved but not exceeded by more than 15% above design at each component; or - group of components operating under a common control in a system containing multiple components, must ensure that each independently operating space of more than 1 000 m² and every separate floor of the building has provision to terminate airflow independently of the remainder of the system sufficient to allow for different operating times; and - must have automatic variable temperature operation of heated water and chilled water circuits; and - when deactivated, must close any motorized outdoor air or return air damper that is not otherwise being actively controlled. When two or more <i>air-conditioning</i> systems serve the same space, they must use control sequences that prevent the systems from operating in opposing heating and cooling modes. Time switches— - A time switch must be provided to control— - an <i>air-conditioning</i> system of more than 2 kW_r; and - a heater of more than 1 kW heating used for <i>air-conditioning</i>. - The time switch must be capable of switching electric power on and off at variable pre-programmed times and on variable pre-programmed days. - The requirements of (i) and (ii) do not apply to— - an <i>air-conditioning</i> system that serves— - only one <i>sole-occupancy unit</i> in a Class 2, 3 or 9c building; or - a Class 4 part of a building; or - a <i>conditioned space</i> where <i>air-conditioning</i> is needed for 24-hour continuous use.

J6D4 - Mechanical ventilation system control	General — A mechanical ventilation system, including one that is part of an <i>air-conditioning</i> system, except where the mechanical system serves only one <i>sole-occupancy unit</i> in a Class 2 building or serves only a Class 4 part of a building, must— - be capable of being deactivated when the building or part of the building served by that system is not occupied; and - when serving a <i>conditioned space</i>, except in periods when evaporative cooling is being used— where specified in Table J6D4, have— - an energy reclaiming system that preconditions outdoor air at a minimum sensible heat transfer effectiveness of 60%; or - demand control ventilation in accordance with AS 1668.2 if appropriate to the application; and - not exceed the minimum outdoor air quantity <i>required</i> by Part F6 by more than 20%, except where— - additional unconditioned outdoor air is supplied for free cooling; or - additional mechanical ventilation is needed to balance the <i>required</i> exhaust or process exhaust; or - an energy reclaiming system preconditions all the outdoor air; and for an airflow of more than 1000 L/s, have a variable speed fan unless the downstream airflow is <i>required</i> by Part F6 to be constant. Exhaust systems — An exhaust system with an air flow rate of more than 1000 L/s must be capable of stopping the motor when the system is not needed, except for an exhaust system in a <i>sole-occupancy unit</i> in a Class 2, 3 or 9c building. <i>Carpark</i> exhaust systems — <i>Carpark</i> exhaust systems must have a control system in accordance with— - Clause 4.11.2 of AS 1668.2; or - Clause 4.11.3 of AS 1668.2. Time switches— A time switch must be provided to a mechanical ventilation system with an air flow rate of more than 1000 L/s. The time switch must be capable of switching electric power on and off at variable pre-programmed times and on variable pre-programmed days. The requirements of (i) and (ii) do not apply to— - a mechanical ventilation system that serves— - only one <i>sole-occupancy unit</i> in a Class 2, 3 or 9c building; or - a Class 4 part of a building; or - a building where mechanical ventilation is needed for 24-hour occupancy.
J6D12 - Unitary air-conditioning equipment	Unitary <i>air-conditioning</i> equipment including packaged air-conditioners, split systems, and variable refrigerant flow systems must comply with <i>MEPS</i> and for a capacity greater than or equal to 65 kW— - where water cooled, have a minimum energy efficiency ratio of 4.0 W_r / W_{input} power for cooling when tested in accordance with AS/NZS 3823.1.2 at test condition T1, where input power includes both compressor and fan input power; or - where air cooled, have a minimum energy efficiency ratio of 2.9 W_r / W_{input} power for cooling when tested in accordance with AS/NZS 3823.1.2 at test condition T1, where input power includes both compressor and fan input power.

Part J7 - Artificial lighting and power

	Required	Compliance Requirement
J7D3 - Artificial lighting	In a sole-occupancy unit of a Class 2 building or a Class 4 part of a building— the lamp power density or illumination power density of artificial lighting must not exceed the allowance of— 5 W/m² within a sole-occupancy unit; and 4 W/m² on a verandah, balcony or the like attached to a sole-occupancy unit; and the illumination power density allowance in (a) may be increased by dividing it by the illumination power density adjustment factor for a control device in Table J7D3b as applicable; and when designing the lamp power density or illumination power density, the power of the proposed installation must be used rather than nominal allowances for exposed batten holders or luminaires; and halogen lamps must be separately switched from fluorescent lamps. In a building other than a sole-occupancy unit of a Class 2 building or a Class 4 part of a building— for artificial lighting, the aggregate design illumination power load must not exceed the sum of the allowances obtained by multiplying the area of each space by the maximum illumination power density in Table J7D3a.	Electrical contractor to ensure the maximum wattage allowances shown above are not exceeded. All artificial lighting and power to comply with Part J7 Refer to Appendix A for calculation
J7D4 - Interior artificial lighting and power control	All artificial lighting of a room or space must be individually operated by— a switch; or other control device; or a combination of (a) and (b).	
J7D5 - Interior decorative and display lighting	Interior decorative and display lighting, such as for a foyer mural or art display, must be controlled— separately from other artificial lighting; and by a manual switch for each area other than when the operating times of the displays are the same in a number of areas such as in a museum, art gallery or the like, in which case they may be combined; and by a time switch in accordance with Specification 40 where the display lighting exceeds 1 kW. Window display lighting must be controlled separately from other display lighting.	
J7D6 - Exterior artificial lighting	Exterior artificial lighting attached to or directed at the facade of a building, must be controlled by— a daylight sensor; or a time switch that is capable of switching on and off electric power to the system at variable pre-programmed times and on variable pre-programmed days	
J7D7 - Boiling water and chilled water storage units	Power supply to a boiling water or chilled water storage unit must be controlled by a time switch in accordance with Specification 40.	
J7D8 - Lifts	Lifts must— be configured to ensure artificial lighting and ventilation in the car are turned off when it is unused for 15 minutes; and achieve the idle and standby energy performance level in Table J7D8a; and achieve the energy efficiency class in Table J7D8b; or if a dedicated goods lift, energy efficiency class D in accordance with ISO 25745-2.	

Part J8 - Heated water supply and swimming pool and spa pool plant

	Required
J8D2 - Heated water supply	A heated water supply system for food preparation and sanitary purposes must be designed and installed in accordance with NCC Volume Three — Plumbing Code of Australia.
J8D3 - Swimming pool heating and pumping	Heating for a swimming pool must be by— - a solar heater; or - a heater using reclaimed heat from another process such as reject heat from a refrigeration plant; or - a geothermal heater; or - a gas heater that— - if rated to consume 500 MJ/hour or less, achieves a minimum gross thermal efficiency of 86%; or - if rated to consume more than 500 MJ/hour, achieves a minimum gross thermal efficiency of 90%; or - a heat pump; or - a combination of (a) to (e). Where some or all of the heating required by (1) is by a gas heater or a heat pump, the swimming pool must have— - a cover with a minimum R-Value of 0.05; and - a time switch to control the operation of the heater. A time switch must be provided to control the operation of a circulation pump for a swimming pool. Where required, a time switch must be capable of switching electric power on and off at variable pre-programmed times and on variable pre-programmed days. Pipework carrying heated or chilled water for a swimming pool must comply with the insulation requirements of J6D9. For the purpose of J8D3, a swimming pool does not include a spa pool.
J8D4 - Spa pool heating and pumping	Heating for a spa pool that shares a water recirculation system with a swimming pool must be by— - a solar heater; or - a heater using reclaimed heat from another process such as reject heat from a refrigeration plant; or - a geothermal heater; or - a gas heater that— - if rated to consume 500 MJ/hour or less, achieves a minimum gross thermal efficiency of 86%; or - if rated to consume more than 500 MJ/hour, achieves a minimum gross thermal efficiency of 90%; or - a heat pump; or - a combination of (a) to (e). Where some or all of the heating required by (1) is by a gas heater or a heat pump, the spa pool must have— - a cover with a minimum R-Value of 0.05; and - a push button and a time switch to control the operation of the heater. A time switch must be provided to control the operation of a circulation pump for a spa pool having a capacity of 680L or more. Where required, a time switch must be capable of switching electric power on and off at variable pre-programmed times and on variable pre-programmed days. Pipework carrying heated or chilled water for a spa pool must comply with the insulation requirements of J6D9.

Part J9 - Energy monitoring and on-site distributed energy resources

	Required
J9D3 - Facilities for energy monitoring	A building or sole-occupancy unit with a floor area of more than 500 m² must have energy meters configured to record the time-of-use consumption of gas and electricity. A building with a floor area of more than 2 500 m² must have energy meters configured to enable individual time-of-use energy data recording, in accordance with (3), of— - air-conditioning plant including, where appropriate, heating plant, cooling plant and air handling fans; and - artificial lighting; and - appliance power; and - central hot water supply; and - internal transport devices including lifts, escalators and moving walkways where there is more than one serving the building; and - on-site renewable energy equipment; and - on-site electric vehicle charging equipment; and - on-site battery systems; and - other ancillary plant. Energy meters required by (2) must be interlinked by a communication system that collates the time-of-use energy data to a single interface monitoring system where it can be stored, analysed and reviewed. The provisions of (2) do not apply to energy meters serving— - a Class 2 building where the total floor area of the common areas is less than 500 m²; or - individual sole-occupancy units with a floor area of less than 2 500 m².

Conclusion

The proposed works have been assessed by NRG efficient homes against Part J of the 2022 NCC Volume 1 and all applicable sections are deemed compliant.

If you have any further questions, feel free to contact me.



Regards
Matthew Morelli
NRG efficient homes
ESD Consultant
DMN Accredited Thermal Performance Assessor
BESS trained professional
Dip. Building Design (Architectural)

DISCLAIMER Although great care has been taken to prepare this report, NRGEH does not make any representations or give any warranties or assurances as to the accuracy of the information contained in the Report or whether the report is free from errors or omissions
NRGEH, its employees and agents shall not be liable for any loss, damage, cost or expense whether direct, indirect or consequential, incurred by, or arising by reason of, any person using or relying on the report and whether caused by reason or error, negligent act, omission or mis representation in the Report or otherwise. The Report has been prepared by NRGEH based on the information supplied.

© Copyright This plan shall remain the sole property of NRGEH and must not be given, lent, hired out or otherwise disposed of or copied without the written permission of NRGEH.

Appendix

Appendix A

Refer to the attached NCC Artificial Lighting Reports

PRELIMINARY

Appendix A



Non-residential Lighting





Building name/description
Workshop - STANWELL POWER STATION, STANWELL

Number of rows preferred in table below: 6 (as currently displayed)

Classification
Class 5

ID	Description	Floor area of the space	Perimeter of the space	Floor to ceiling height	Design illumination power load	Space	Illuminance		Adjustment factor 1		Adjustment factor 2		Light colour adjustment factors		SATISFIES PART 7JD3	
							Designed lux level	Recommended lux level	Adjustment factor 1	Dimming % area	Illuminance turn-down	Adjustment factor 2	Dimming % area	Illuminance turn-down	Light colour adjustment factor 1	Light colour adjustment factor 2
1	Open Office	400.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more							gCCT = 4000 K	gCRI = 90	180 W	100% W
2	Large Meeting	20.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									171 W	100% W
3	Small Meeting	15.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									88 W	100% W
4	Open Office	15.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									23 W	100% W
5	Reception	55.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									23 W	100% W
6	Plant Room	55.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									220 W	100% W
7	Male	47.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									141 W	100% W
8	Female	50.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									0 W	100% W
9	Female	18.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									54 W	100% W
10	Clean	2.2 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									3 W	100% W
11	AL	12.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									36 W	100% W
12	Lunch Room	75.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									225 W	100% W
															Total	1568 W

If inputs are valid 

IMPORTANT NOTICE AND DISCLAIMER IN RESPECT OF THIS LIGHTING CALCULATOR
The calculator is provided as a guide only and does not constitute a recommendation or warranty. It is the user's responsibility to ensure that the calculator is used correctly and that the results are used in accordance with the relevant standards and regulations. The calculator is not intended to be used for any other purpose than the calculation of lighting requirements. The calculator is not intended to be used for any other purpose than the calculation of lighting requirements. The calculator is not intended to be used for any other purpose than the calculation of lighting requirements.



Non-residential Lighting





Building name/description
Workshop - STANWELL POWER STATION, STANWELL

Number of rows preferred in table below: 12 (as currently displayed)

Classification
Class 5

ID	Description	Floor area of the space	Perimeter of the space	Floor to ceiling height	Design illumination power load	Space	Illuminance		Adjustment factor 1		Adjustment factor 2		Light colour adjustment factors		SATISFIES PART 7JD3	
							Designed lux level	Recommended lux level	Adjustment factor 1	Dimming % area	Illuminance turn-down	Adjustment factor 2	Dimming % area	Illuminance turn-down	Light colour adjustment factor 1	Light colour adjustment factor 2
1	Open Office	400.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more								gCCT = 4000 K	gCRI = 90	180 W
2	Large Meeting	20.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									171 W	
3	Small Meeting	15.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									88 W	
4	Open Office	15.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									23 W	
5	Reception	55.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									23 W	
6	Plant Room	55.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									220 W	
7	Male	47.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									141 W	
8	Female	50.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									0 W	
9	Female	18.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									54 W	
10	Clean	2.2 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									3 W	
11	AL	12.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									36 W	
12	Lunch Room	75.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									225 W	
															Total	2813 W

If inputs are valid 

IMPORTANT NOTICE AND DISCLAIMER IN RESPECT OF THIS LIGHTING CALCULATOR
The calculator is provided as a guide only and does not constitute a recommendation or warranty. It is the user's responsibility to ensure that the calculator is used correctly and that the results are used in accordance with the relevant standards and regulations. The calculator is not intended to be used for any other purpose than the calculation of lighting requirements. The calculator is not intended to be used for any other purpose than the calculation of lighting requirements. The calculator is not intended to be used for any other purpose than the calculation of lighting requirements.



Non-residential Lighting





Building name/description
Main Warehouse - STANWELL POWER STATION, STANWELL

Number of rows preferred in table below: 13 (as currently displayed)

Classification
Class 5

ID	Description	Floor area of the space	Perimeter of the space	Floor to ceiling height	Design illumination power load	Space	Illuminance		Adjustment factor 1		Adjustment factor 2		Light colour adjustment factors		SATISFIES PART 7JD3	
							Designed lux level	Recommended lux level	Adjustment factor 1	Dimming % area	Illuminance turn-down	Adjustment factor 2	Dimming % area	Illuminance turn-down	Light colour adjustment factor 1	Light colour adjustment factor 2
1	Open Office	400.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more								gCCT = 4000 K	gCRI = 90	180 W
2	Large Meeting	20.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									171 W	
3	Small Meeting	15.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									88 W	
4	Open Office	15.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									23 W	
5	Reception	55.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									23 W	
6	Plant Room	55.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									220 W	
7	Male	47.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									141 W	
8	Female	50.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									0 W	
9	Female	18.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									54 W	
10	Clean	2.2 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									3 W	
11	AL	12.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									36 W	
12	Lunch Room	75.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									225 W	
13	Small Meeting	13.0 m²		3.0 m	10 W/m²	Office - additional to an adjacent level of 200 lux or more									59 W	
															Total	1278 W

If inputs are valid 

IMPORTANT NOTICE AND DISCLAIMER IN RESPECT OF THIS LIGHTING CALCULATOR
The calculator is provided as a guide only and does not constitute a recommendation or warranty. It is the user's responsibility to ensure that the calculator is used correctly and that the results are used in accordance with the relevant standards and regulations. The calculator is not intended to be used for any other purpose than the calculation of lighting requirements. The calculator is not intended to be used for any other purpose than the calculation of lighting requirements. The calculator is not intended to be used for any other purpose than the calculation of lighting requirements.